Tyler Dinh

816-905-6348 | tylerdinh14@gmail.com | linkedin.com/in/dinhtyler | github.com/tylerdinh

Education

Washington University in St. Louis — St. Louis, MO
Bachelor of Science in Computer Science; Minor in Linguistics — *Aug. 2024 – May 2028*
- GPA: 3.78/4.0

Experience

Amazon — Arlington, VA
Software Development Engineer Intern — *May 2025 – Aug. 2025*
- Engineered foundational vision-language models and data processing pipelines for real-time sports broadcast analysis, scaling content coverage capacity from 4,000 to 250,000 instances (60x increase) for Amazon Prime Video Sports, creating the first automated, AI-augmented broadcast state detector within the company.
- Designed pub-sub and Lambda microservice architecture in AWS to process live content analysis with sub-5 second latency, successfully meeting director-provided service-level agreements for Prime Video Sports operations.
- Established multiple AI Safety guidelines and wrote comprehensive service documentation for stakeholder alignment on critical service decisions, providing guidance on how to effectively modify systems within the service.

WashU Robotics — St. Louis, MO
Nimble Project Lead and Software Lead — *Aug. 2024 – Present*
- Trained industrial articulated robot arm using the OpenVLA-7B foundational vision-language-action model with reinforcement learning and fine-tuning techniques in Python to achieve out-of-distribution generalization and long-sequence planning on human manipulation tasks.
- Researched tradeoffs of foundational vision-language models for robotic use-cases such as writing a sentence, designing a lossless protocol for transfer between a vision-language model and vision-language-action model.
- Led and recruited a 15-member team within 200+ member organization, overseeing project roadmap, deliverables, and budget allocation to drive team-level success and growth.

FIRST Robotics Competition Team 5013 — Kansas City, MO
Team Captain and Software Lead — *Aug. 2020 – May 2024*
- Developed TensorFlow-based object recognition system with bounding box detection in Python to improve autonomous routine performance for command-based Java code.
- Optimized PID control systems for precision targeting, implementing linear interpolation to calculate optimal shot angles based on field position.

Projects

Campus Creamery Mobile App | *React Native, Firebase, Express, Node.js, Git* — Aug. 2024 – Aug. 2025
- Built a full-stack mobile application for a student-run ice cream store under WashU Google Developer Group, implementing agile workflows and comprehensive project documentation to efficiently manage the project.

Honors & Awards

WashU Academic Dean's List — Fall 2024, Spring 2025
Advanced Placement Scholar with Distinction (16 AP Courses) — July 2024
Congressional App Challenge Winner (Project recognized at U.S. Capitol) — Apr. 2024
Amazon Future Engineer Scholarship (1 of 400 worldwide recipients) — Mar. 2024
National Merit Commended Scholar — Sep. 2023
FIRST Robotics Dean's List Finalist (Top 300 of 85,000+ participants) — Mar. 2023
FIRST Robotics World Championship Qualifier (2x) — Apr. 2022, Apr. 2023

Technical Skills

Languages: Java, Python, TypeScript, JavaScript, PHP, HTML, CSS, SQL, C++, Arduino C
Frameworks & Libraries: React, React Native, TensorFlow, pandas, NumPy, Matplotlib, JUnit
Cloud & DevOps: AWS (Lambda, SNS, SQS, Bedrock, EC2, CloudWatch), Git, GitHub, Linux, Terminal/CLI, CI/CD
Areas of Interest: AI Safety, Machine Learning, Computational Linguistics, Robotics, Product Development